Thermodynamic description of the Cr–Fe system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Cr–Fe binary system as calculated with pycalphad from the SGTE dataset CrFe-11Xio.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, σ → (Cr,Fe)1 + (Cr,Fe)2, lies at 790 K.

1. Introduction

The Cr–Fe system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic, gamma loop, solid miscibility gap. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of W. Xiong, P. Hedstrom, M. Selleby, J. Odqvist, M. Thuvander, Q. Chen, Calphad, 35, 355-66(2011). The calculations use pycalphad 0.11.2 at 1 atm.

Pure Cr melts at 2180 K. For Fe, the liquidus meets the axis at 1818 K.

2. Phase diagram

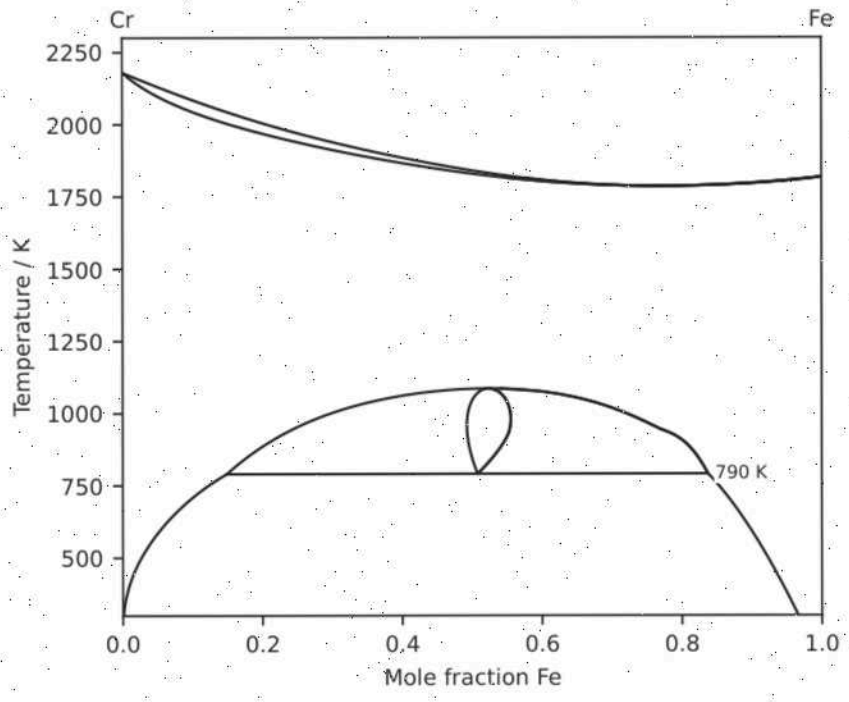


Fig. 1. Calculated Cr–Fe phase diagram.

The dataset places a eutectoid (σ → (Cr,Fe)1 + (Cr,Fe)2) at 790 K; the composition of σ is 50.8 at.% Fe.

Table 1. Invariant reactions in the Cr–Fe system (calculated).

Reaction	Type	T / °C	Compositions (x(Fe))		
			phase 1	phase 2	phase 3
σ → (Cr,Fe)1 + (Cr,Fe)2	eutectoid	517.0	0.508	0.149	0.837

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 2250 K the integral mixing enthalpy of the liquid has its maximum, 0.04 kJ/mol, at 78.0 at.% Fe. The calculated enthalpy of formation of σ at 298.15 K is 5.89 kJ/mol per mole of atoms.

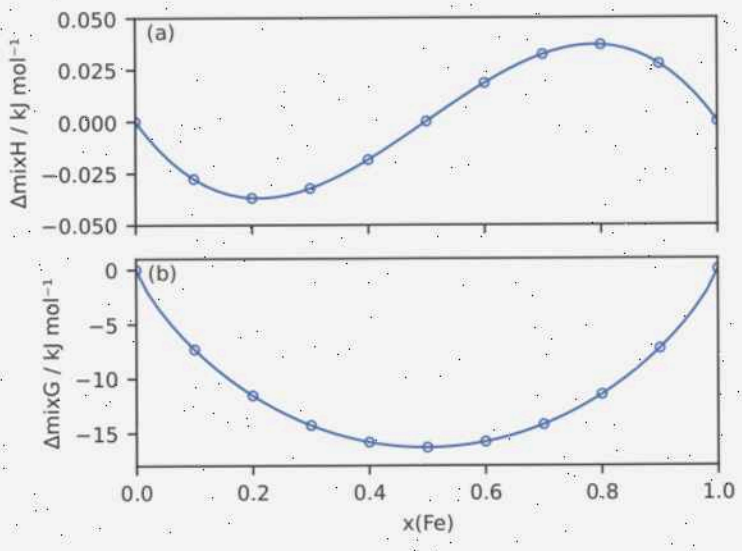


Fig. 2. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 2250 K.

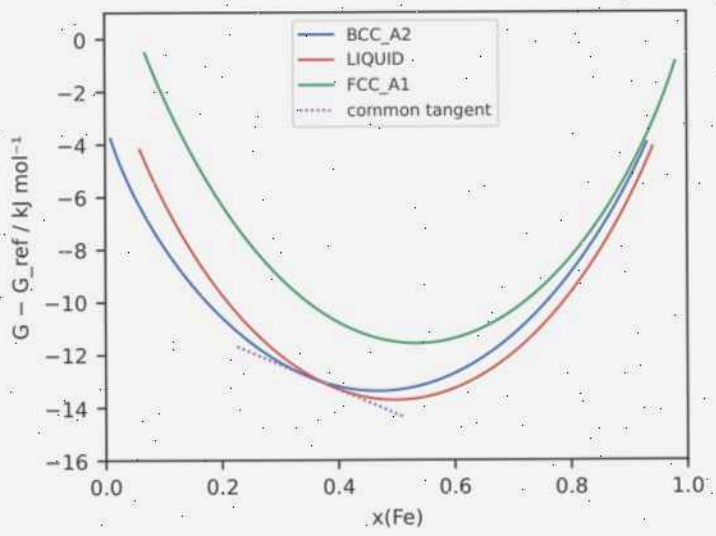


Fig. 3. Calculated Gibbs energy curves at 1890 K.

Table 2. Calculated formation properties of the intermediate phases (Cr–Fe).

		Calculated at 298.15 K, per mole of atoms	
Phase	**at.% Fe**	**ΔfH / kJ mol⁻¹**	**S / J mol⁻¹ K⁻¹**
σ	52.0	5.89 ± 2.0	34.56

Reference: stable elements (SER) at 298.15 K. ± values are synthetic illustration uncertainties, not data.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.